

Lorenzo Loyola

Civil Engineer, MSc – PhD candidate (applicant)
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Education

- M.Sc. in Civil and Architectural Engineering, Aarhus University, Denmark (2025). Thesis: “Digital Twin and Virtual Environment for Autonomous Bridge Inspection”, grade 12/12.
- Civil Engineer, Structural and Construction (professional degree), University of Chile (2018). Thesis: “Development of a BIM software application for interoperability between structural analysis and 3D reinforcement representation in reinforced-concrete walls”, grade 7.0/7.0.
- Bachelor of Engineering Sciences, University of Chile (2014).

Employments

- Research Assistant, Department of Civil and Architectural Engineering, Aarhus University, Denmark (2026-now): ontology and knowledge graph for structural monitoring data.
- Research Assistant (project-based), Aarhus University, Denmark (2025): real-time CO₂-monitoring digital twin for a construction site with Acembee, Clevertrack and NIRAS.
- Student Assistant, Aarhus University, Denmark (2024): national survey on Lean construction and health and safety.
- R&D Senior Consultant, Rene Lagos Engineers, Chile (2026-now, part-time, remote).
- Head of R&D, Rene Lagos Engineers, Chile (2021-2026): led a team of four engineers developing BIM automation tools.
- R&D Structural Engineer, Rene Lagos Engineers, Chile (2017-2021): reinforced-concrete design, structural instrumentation and monitoring, post-earthquake assessment.
- Assistant Engineer, Technical Office, Chile (2016-2017).

Research achievements and impact

Lorenzo Loyola’s research combines structural-engineering practice with digital construction. In industry he led the development of Revit add-ins that automate the transfer of structural design data into BIM reinforcement models, and carried out accelerometer-based monitoring and post-earthquake field assessments. At Aarhus University he deployed a real-time CO₂-monitoring digital twin on an active construction site with three industry partners (two first-author papers), built a BIM-based digital twin coupled to a Gazebo/ROS 2 virtual environment for simulated autonomous UAV bridge inspection (MSc thesis, submitted to i3CE 2026), and currently develops an ontology, Neo4j knowledge graph and LLM interface for structural monitoring data, the foundation of WP1 and WP2 of this proposal. Training: Urban Digital Twin PhD Spring School, London (2026); NRIS course on distributed LLM fine-tuning on HPC (2026).

Supervision record

Led a team of four engineers in industry (2021-2026).

Technical skills

Structural analysis (ETABS, SAFE, Robot, OpenSees, MATLAB); BIM automation (Revit API/C#, Dynamo, Rhino/Grasshopper, IFC); data and AI (Python, SQL, Neo4j, LLM APIs, LangChain, scikit-learn); simulation and robotics (ROS 2, Gazebo, ArduPilot, MAVLink, MAVSDK). Languages: Spanish (native), English (full professional), Danish (A2).

Selected publications

Five most relevant to the project:

1. Loyola, L., Martinez, J. G., & Abbiati, G. (2026). BIM-based digital twin for UAV bridge inspection. ASCE International Conference on Computing in Civil Engineering (i3CE 2026), submitted.
2. Loyola, L., et al. (in preparation). An ontology for structural monitoring data. Journal article.
3. Singh, S., et al., incl. Loyola, L. (2026). LLM-assisted chatbot for data-driven construction monitoring and decision support. AUC-CSCE Special Issue 50.
4. Loyola, L. I., & Martinez, J. G. (2025). Real-time energy and CO₂ optimization in construction cranes through digital twins and lean construction. Revista Ingeniería de Construcción, [10.7764/RIC.00166.21](#).
5. Loyola, L. (2025). Digital Twin and Virtual Environment for Autonomous Bridge Inspection. MSc thesis, Aarhus University.

Other recent publications (last 5 years):

1. Martinez Ribon, J. G., Montanari, P., Loyola, L. I., & Singhal, S. (2025). Uso de gemelos digitales y sensores IoT para la eficiencia energética de grúas en construcción. Simpósio Brasileiro de Tecnologia da Informação e Comunicação na Construção (SBTIC 2025), [10.46421/sbtic.v5i00.8211](#).
2. Ullah, A. N. L., Puja, T. C., Martinez, J., Loyola, L., & Perez, F. (2026). Proactive safety management in earthwork operations using IoT-driven real-time risk analysis. European Conference on Computing in Construction (EC3 2026).
3. Anam, et al., incl. Loyola, L. (2026). Requests for information in construction projects: comparing systematic literature evidence with industry survey findings. CIB W78 2026, New Delhi, India.
4. Pérez, C. T., Madushanka, M., Loyola, L., Ergul, M., & Salling, S. T. (2025). Health and safety perceptions of construction workers in Denmark. 23rd CIB World Building Congress, West Lafayette, USA, [10.7771/3067-4883.1125](#).
5. Pérez, C. T., Madushanka, M., Loyola, L., Ergul, M., Salling, S. T., & Wandahl, S. (2024). The impact of Lean knowledge and Lean operation on construction workers' job satisfaction. 32nd Annual Conference of the International Group for Lean Construction (IGLC32), [10.24928/2024/0188](#).